Chang Zeng

Senior Research Scientist | Generative Audio & Multimodal AI

Tokyo, Japan | (+86)13821557429 | zengchang.elec@gmail.com

| github.com/zengchang233 | scholar.google.com/citations?user=gfGyn49j-MkC&hl=en

Senior Research Scientist (Audio/Multimodal AI) Ph.D. in Informatics with **7+ years of experience** bridging cutting-edge research and industrial-scale deployment in **Generative Audio** and **Spoken Language Models**. Currently leading a research team to build **full-duplex voice agents** from scratch, covering the entire pipeline from large-scale training (GPT-OSS/MoE) to Codec optimization. Proven track record in **TTS**, **Singing Voice Synthesis**, and **ASR**, with 10+ top-tier publications (ICLR, Interspeech, ICASSP). Expert in translating theoretical breakthroughs (GANs, Diffusion, Transformers) into practical, product-ready creative tools. Passionate about redefining content creation through intuitive, high-quality audio-visual AI.

Education

National Institute of Informatics (SOKENDAI) **2020.10 - 2024.03.22**

Doctor of Informatics

Sponsored by the Japanese government MEXT scholarship, supervised by Prof. Yamagishi.

The University of Tokyo **2017.10 - 2020.03**

Master of Electrical Engineering and Information System

Supervised by Prof. Minematsu.

Tianjin University **2012.09 - 2016.07**

Bachelor of Engineering

Publications

Featured Publications

1. **Zeng, C**., & Wang, C. (2024). InstructSing: High-fidelity Singing Voice Generation via Instructing Yourself. *IEEE SLT 2024*. Demo page.
2. **Zeng, C**., Miao, X., Wang, X., Cooper, E., & Yamagishi, J. (2024). Spoofing-Aware Speaker Verification Robust Against Domain and Channel Mismatches. *IEEE SLT 2024*.
3. **Zeng, C**., Miao, X., Wang, X., Cooper, E., & Yamagishi, J. (2024). Joint Speaker Encoder and Neural Back-end Model for Fully End-to-End Automatic Speaker Verification with Multiple Enrollment Utterances. *Journal of Computer Speech & Language. (JCR Q1)*
4. **Zeng, C**., Wang, X., Miao, X., Cooper, E., & Yamagishi, J. (2023). Improving Generalization Ability of Countermeasures for New Mismatch Scenario by Combining Multiple Advanced Regularization Terms. In *Proc. Interspeech 2023, 1998-2002. (CoreRank A)*
5. **Zeng, C**., Wang, X., Cooper, E., Miao, X., & Yamagishi, J. (2022). Attention back-end for automatic speaker verification with multiple enrollment utterances. In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 6717-6721). Project page. *(CoreRank B)*
6. **Zeng, C**., Zhang, L., Liu, M., & Yamagishi, J. (2022). Spoofing-Aware Attention based ASV Back-end with Multiple Enrollment Utterances and a Sampling Strategy for the SASV Challenge 2022. In *Proc. Interspeech 2022, 2883–2887. (CoreRank A)*
7. Wang, C., **Zeng, C (co-first author)**., & He, X. (2023). Xiaoicesing 2: A High-Fidelity Singing Voice Synthesizer Based on Generative Adversarial Network. In *Proc. Interspeech 2023, 5401-5405*. Project page. *(CoreRank A)*
8. Wang X., **Zeng, C (co-first author)**., Chen, J., Wang, C. (2023). CrossSinger: A Cross-Lingual Multi-Singer High-Fidelity Singing Voice Synthesizer Trained on Monolingual Singers. *IEEE ASRU 2023*. Project page.
9. Li, K., Sang, W., **Zeng, C.**, Yang, R., Chen, G., Hu, X. (2024). LibriSpace: A Simulated Moving Audio Dataset for Speech Enhancement and Separation. *ICLR 2025*.
10. Wang, C., **Zeng, C (Co-first author)**., Zhang, B., Ma, Z., Zhu, Y., Cai, Z., Zhao, J., Jiang, Z., & Chen, Y. (2024). HAM-TTS: Hierarchical Acoustic Modeling for Token-Based Zero-Shot Text-to-Speech with Model and Data Scaling. Demo Page.

Work Experiences

Shanda AI Research Tokyo **2025.09 - Present**

Senior AI Researcher

Responsibilities

- Responsible for the research and development of full-duplex dialogue agent for Shanda's Avatar and Game products.

Achievements

- Led a 4-person team to build a full-duplex conversational voice agent system from scratch based on a GPTOSS-20B-A3.4B MoE model, providing voice-driven services for Shanda's virtual human, world model, and gaming business.
- Engaged in the end-to-end development pipeline, including data cleaning, Codec model design and training, construction of multi-machine multi-GPU LLM training pipelines, and performance evaluation.
- Achieved state-of-the-art (SOTA) surpassing objective metrics (e.g., PESQ) with the developed Codec model under 1Kbps conditions, enabling full-duplex voice dialogue capabilities based on a single model.

- Explored proactive dialogue capabilities, empowering the voice agent (especially game NPCs) to autonomously perceive environmental changes and initiate conversations, facilitating self-evolution and knowledge base updates.
- Acted as Co-PI, co-leading a joint research project with Prof. Hu (TSAIL, Tsinghua University) and his Ph.D. Candidate Kai Li to investigate LLM-based speech interaction models for cocktail party scenarios. Delivered an initial-stage outcome: a universal, LLM-based data processing pipeline and model architecture for audio separation.

Li Auto **2024.04 - 2025.09**

Multimodal Generative Researcher

Responsibilities

- Responsible for the research and development of voice generation for Li Auto's smart space, as well as the R&D of MindGPT-4o, Li Auto's multimodal foundation large model.

Achievements

- Proposed the GFSQ (Group Finite Scalar Vector Quantization) audio tokenizer for GPT-SoVITS modeling. This method effectively enhances the codebook utilization of standard RVQ and improves GPT decoding accuracy through the use of delay pattern decoding.
- Trained a multi-timbre and multi-style model for Li Auto's voice blog service. Utilized an internal foundational TTS large model to generate training data with specific styles and timbres, fine-tuning GPT-SoVITS to provide a selection of high-quality styles and voices for the blog service.
- Managed data production and audio head training for the MindGPT-4o full-duplex conversational large model. Replicated a diverse range of data styles using the internal foundational TTS large model, including but not limited to various accents, dialects, foreign languages, emotions, and scenarios. Also responsible for the pre-training and post-training tasks of the audio head to expand its customization capabilities.

Xiaoice **2022.07 - 2023.07**

Avatar Research Intern

Responsibilities

- I am focusing on building a generative model for high-fidelity (48kHz) singing voice generation tasks collaborating with other researchers and engineers.

Achievements

- Improved the Fastspeech2-based XiaoiceSing to **XiaoiceSing2** by utilizing **adversarial training** to generate a mel-spectrogram with high accuracy. The MOS score shows that the quality of synthesized singing voices is quite close to the human level (4.23 vs 4.27). The paper has been accepted by Interspeech 2023;
- Developed a GAN-based vocoder model called **HiFi-WaveGAN** to reconstruct the waveform from the mel-spectrogram by incorporating a novel **Pulse Sequence**. The proposed vocoder beats the widely-used HiFi-GAN model in terms of MOS metric for 48kHz singing voice synthesis task (4.23 vs 3.89);
- Improved XiaoiceSing2 to a cross-lingual (en, jp, cn) multi-singer singing voice synthesizer **CrossSinger** by incorporating language information into the encoder and leveraging a novel **bias eliminator** to remove singer information contained in the encoder. The paper has been accepted by ASRU 2023;
- Improved HiFi-WaveGAN to **InstructSing** by fusing differential digital signal processing (DDSP) and adversarial training. The imperative information generated by the DDSP module is refined by a UNet-based module to instruct and accelerate the speed of the adversarial training. Experiments show that the proposed model can converge within 20,000 training steps, which is **one-tenth** of other neural vocoders.
- Hierarchical modeling for token-based large voice language model. Hierarchical modeling acoustic context information by inserting the HuBert Unit as the intermediate representation for the text module of VALLE. The pronunciation accuracy of synthesized speech is significantly improved.

ALIBABA **2020.04 - 2020.09**

Speech Recognition Researcher (Full-time)

Responsibilities

- In Alibaba, I belonged to a department that is responsible for TAOBAO living. To prevent live broadcasters from violating laws and regulations, I developed two systems with my colleagues.

Achievements

- Developed a large-scale speaker recognition system aiming at identifying the talker in each living room as the exact person;
- Developed a spoken word detection system aims at catching some illegal words spoken by broadcasters;
- Explored SSL for speech representation and developed an end-to-end speech recognition system with knowledge distillation based on ESPNet.

Activities

Competitions

- 4th/77 place for VoxCeleb Speaker Recognition Challenge 2019.
- 2nd/110 place for Zhijiang Cup Speech Recognition for Conversational Scenario 2021.
- 4th/42 place for Audio Deep Synthesis Detection Challenge 2022 track1 (Low-quality Fake Audio Detection, LF).

- 5th/27 place for Audio Deep Synthesis Detection Challenge 2022 track2 (Partially Fake Audio Detection, PF).

Academic Activities

- Research assistant at Yamagishi Lab, National Institute of Informatics
- ICASSP 2022 Oral Presentation.
- Interspeech 2022 Oral Presentation.
- Interspeech 2023 Poster Presentation.
- Reviewer of IEEE Open Journal of Signal Processing.
- Reviewer of ICASSP, ICME, Interspeech, NeurIPS, ICLR, ICML, and AISTATS.
- Organization committee of Joint Workshop of VoicePersonae and ASVspoof 2023.

Open Source

- WeSpeaker contributor (660 stars), ASV-Subtools contributor (590 stars), Attention backend

Skills

Tools and Languages	Python, CPP, Shell, Git, MySQL
Toolkits	PyTorch, Pytorch-Lightning, SpeechBrain, WeNet, WeSpeaker, Kaldi, Espnet, HuggingFace
Communication	Chinese (native), English (business level), Japanese (N2-125)